1st International Workshop on

Quantum Intelligence, Security, and Networking

In conjunction with the 36th ICCCN | Rome, Italy | Workshop date: TBA

Workshop Scope

The 1st International Workshop on Quantum Intelligence, Security, and Networking (QISNet 2027) aims to provide an interdisciplinary forum for researchers and practitioners exploring how quantum science and technology can transform future communication, networking, distributed computing, artificial intelligence, and cybersecurity.

QISNet 2027 focuses on the convergence of quantum computing, quantum artificial intelligence, quantum communications, and quantum security with next-generation computer networks and distributed systems. Quantum computing and quantum machine learning may enable new approaches to network optimization, resource allocation, traffic analysis, distributed intelligence, and complex decision-making.

Quantum communication technologies - including quantum key distribution, entanglement distribution, quantum repeaters, and emerging quantum-network architectures - introduce fundamentally new models for secure and distributed communication. At the same time, advances in quantum computing motivate a transition toward post-quantum cryptography, crypto-agility, and quantum-resilient network infrastructures.

The workshop places a strong emphasis on communications and networking, including architectures, protocols, algorithms, security mechanisms, intelligent resource-management methods, experimental systems, testbeds, and applications that connect quantum technologies with computer communications and networks. The workshop welcomes theoretical, experimental, and systems-oriented work, including hybrid quantum-classical approaches.

Topics of Interest

Topics include, but are not limited to:

Quantum computing & intelligence

- Quantum computing for communication and network optimization
- Quantum algorithms for routing, scheduling, resource allocation, and network design
- Quantum machine learning and quantum AI for networks
- Hybrid quantum-classical AI for networked and distributed systems
- Quantum-enhanced network traffic analysis, prediction, and anomaly detection
- Quantum-enabled optimization for edge, cloud, and distributed computing
- Quantum-enhanced distributed computing and federated or collaborative intelligence

Quantum communication & networking

- Quantum communication networks and architectures
- Quantum internet architectures, protocols, and services
- Entanglement generation, distribution, routing, and resource management
- Quantum repeaters, quantum memories, and network control
- Classical-quantum network integration and control planes
- Quantum networking for data centers, edge systems, IoT, and cyber-physical systems

Topics of Interest (continued)

Security & quantum resilience

- Quantum key distribution networks and secure key management
- Post-quantum cryptography for communication and network systems
- Crypto-agility and migration to quantum-safe network infrastructures
- Quantum-resilient authentication, identity, and trust management
- Secure protocol design for quantum and post-quantum networks
- Quantum threat modeling, risk assessment, and network security
- AI-assisted security for quantum and post-quantum systems

Systems, testbeds & applications

- Quantum network simulation, modeling, emulation, and testbeds
- Quantum sensing and networked quantum sensors
- Resource management and performance evaluation in quantum-enabled networks
- Reliability, scalability, interoperability, and standardization of quantum networks
- Experimental platforms, prototypes, applications, and use cases for quantum-enabled networks

Submission Guidelines

Authors are invited to submit original, unpublished research. Include an abstract, keywords, and the email address of the corresponding author. Papers should be limited to **6 pages** in standard IEEE camera-ready format (**double-column, 10-point font**).

The submission link and detailed ICCCN 2027 instructions will be posted when available. Final requirements will follow the ICCCN 2027 workshop submission, formatting, registration, and presentation policies.

Author commitment: If a paper is accepted, at least one author must register for and attend the workshop to present the work.

Review and Publication

All submitted papers will be peer reviewed. Accepted papers must satisfy the final ICCCN 2027 registration, presentation, formatting, and publication requirements.

Accepted and presented workshop papers are expected to be included in the ICCCN conference proceedings and submitted for publication in IEEE Xplore, subject to the policies and technical co-sponsorship arrangements announced by ICCCN 2027.

Important Dates

Paper submission: **TBA**

Acceptance notification: **TBA**

Camera-ready deadline: **TBA**

Workshop date: **TBA**

Dates will be aligned with ICCCN 2027.

Workshop Chairs & Contact

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